



Credit Card Customer Segmentation

UNSUPERVISED LEARNING



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ANNAPOORNA S

Credit Card Customer Segmentation

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Problem Statement

Context

AllLife Bank wants to focus on its credit card customer base in the next financial year. They have been advised by their marketing research team, that the penetration in the market can be improved. Based on this input, the Marketing team proposes to run personalized campaigns to target new customers as well as upsell to existing customers. Another insight from the market research was that the customers perceive the support services of the bank poorly. Based on this, the Operations team wants to upgrade the service delivery model, to ensure that customer queries are resolved faster. The Head of Marketing and Head of Delivery both decide to reach out to the Data Science team for help

Objective

To identify different segments in the existing customers, based on their spending patterns as well as past interaction with the bank, using clustering algorithms, and provide recommendations to the bank on how to better market to and service these customers.

Data Description

The data provided is of various customers of a bank and their financial attributes like credit limit, the total number of credit cards the customer has, and different channels through which customers have contacted the bank for any queries (including visiting the bank, online, and through a call center).

Data Dictionary

- SL_No: Primary key of the records
- Customer Key: Customer identification number
- Average Credit Limit: Average credit limit of each customer for all credit cards
- Total credit cards: Total number of credit cards possessed by the customer
- Total visits bank: Total number of visits that the customer made (yearly) personally to the bank
- Total visits online: Total number of visits or online logins made by the customer (yearly)

- Total calls made: Total number of calls made by the customer to the bank or its customer service department (yearly)

Exploratory Data Analysis

Data Overview

- observe the first few rows of the dataset, to check whether the dataset has been loaded properly or not
- get information about the number of rows and columns in the dataset
- find out the data types of the columns to ensure that data is stored in the preferred format and the value of each property is as expected.
- check the statistical summary of the dataset to get an overview of the numerical columns of the data

	Sl_No	Custo mer Key	Avg_Credi t_Limit	Total_Credit _Cards	Total_visit s_bank	Total_visits _online	Total_calls _made
0	1	87073	100000	2	1	1	0
1	2	38414	50000	3	0	10	9
2	3	17341	50000	7	1	3	4
3	4	40496	30000	5	1	1	4
4	5	47437	100000	6	0	12	3

Table 1: First 5 rows of the data set

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 660 entries, 0 to 659
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   Sl_No                 660 non-null    int64
1   Customer Key          660 non-null    int64
2   Avg_Credit_Limit      660 non-null    int64
3   Total_Credit_Cards    660 non-null    int64
4   Total_visits_bank     660 non-null    int64
5   Total_visits_online   660 non-null    int64
6   Total_calls_made      660 non-null    int64
dtypes: int64(7)
memory usage: 36.2 KB

```

Figure 1: Data information with data types

- The data frame contains 660 rows and 7 columns.
- All columns have integer data type (int64).
- There are no missing values in any column
- The columns in the dataset are:
 - *Sl_No*: Primary key/row identifier
 - *Customer Key*: Customer identification number
 - *Avg_Credit_Limit*: Average credit limit of each customer
 - *Total_Credit_Cards*: Total number of credit cards possessed by the customer
 - *Total_visits_bank*: Total yearly visits to the bank in person
 - *Total_visits_online*: Total yearly online logins
 - *Total_calls_made*: Total yearly calls to customer service
- Memory usage is relatively small at 36.2 KB.

Statistical summary of data set

	count	mean	std	min	25%	50%	75%	max
Sl_No	660.0	330.500000	190.669872	1.0	165.75	330.5	495.25	660.0
Customer Key	660.0	55141.443939	25627.772200	11265.0	33825.25	53874.5	77202.50	99843.0
Avg_Credit_L	660	34574.24	37625.48	3000	10000	1800	48000	20000

	count	mean	std	min	25%	50%	75%	max
limit	.0	2424	7804	.0	.00	0.0	.00	0.0
Total_Credit_Cards	660 .0	4.706061	2.167835	1.0	3.00	5.0	6.00	10.0
Total_visits_bank	660 .0	2.403030	1.631813	0.0	1.00	2.0	4.00	5.0
Total_visits_online	660 .0	2.606061	2.935724	0.0	1.00	2.0	4.00	15.0
Total_calls_made	660 .0	3.583333	2.865317	0.0	1.00	3.0	5.00	10.0

Table 2: Statistical Summary of data set

Distribution Insights:

- Total_visits_online and Avg_Credit_Limit have high positive skew (2.23 and 2.20)
- This indicates that most customers have fewer online visits and lower credit limits, with a few customers having much higher values.
- Total_calls_made has moderate skewness (0.66), suggesting some customers require more support than others.
- Total_Credit_Cards and Total_visits_bank have low skewness (0.14 and 0.14), indicating more balanced distributions.

Customer Profile Insights:

- Average credit limit: \$34,574.24, with maximum reaching \$200,000.00
- Customers have an average of 4.7 credit cards
- Customers visit online 2.6 times per year on average
- Customers make 3.6 calls to the bank per year on average

Potential Segmentation Factors:

- Credit limit and number of cards could identify premium vs standard customers
- Online visits vs physical visits may indicate digital preference
- Number of calls may indicate service needs or issues with current services

Marketing & Service Implications:

- High skewness in credit limits suggests opportunity for tier-based marketing
- Variation in channel preferences (online vs in-person) suggests need for channel-specific strategies
- Call volume data can help optimize customer service staffing and identify frequent pain points

Univariate Analysis

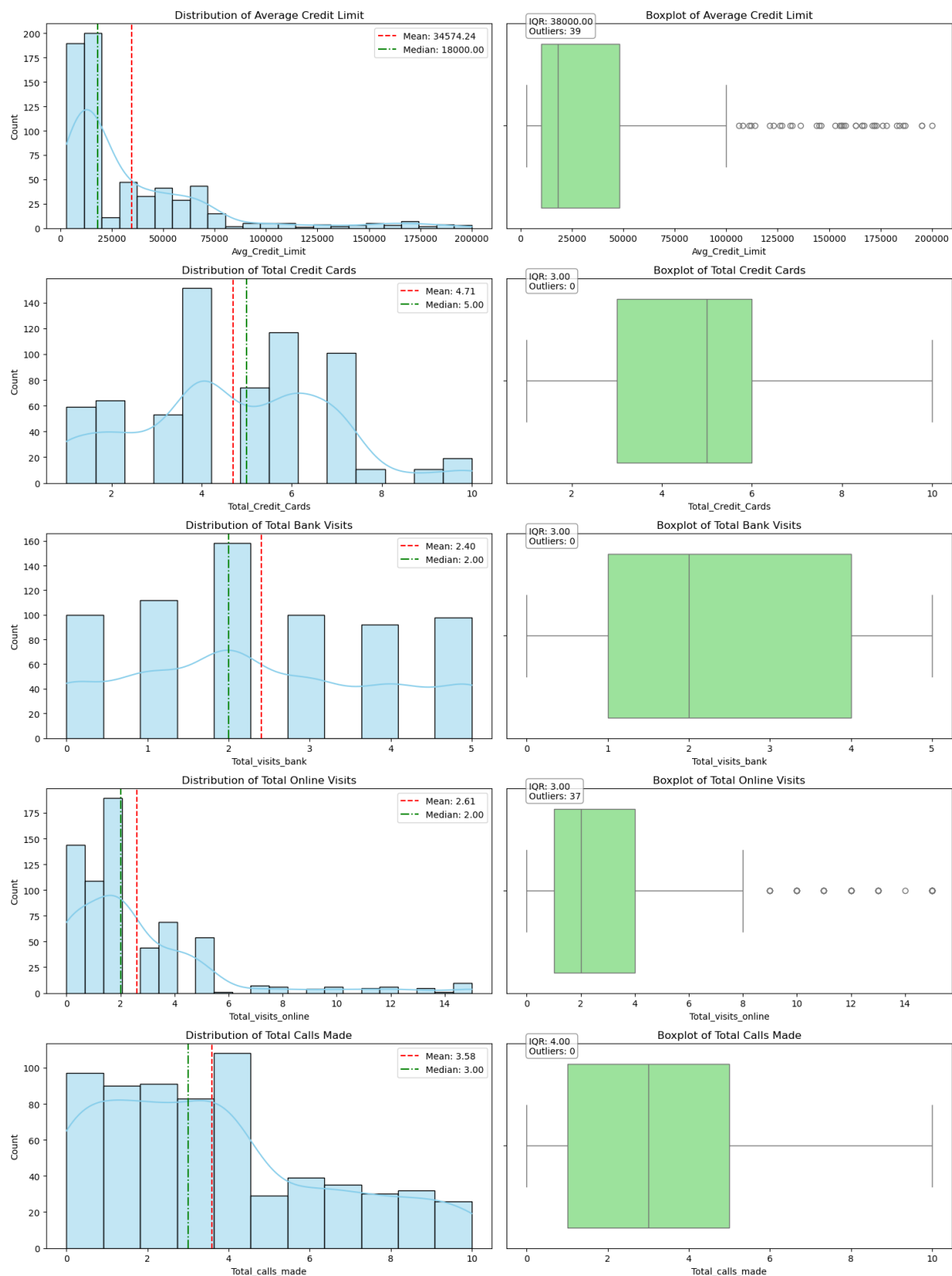


Figure 2: Univariate analysis of each distribution using histogram and box plot

Average Credit Limit

- The average credit limit is approximately \$34,574, significantly higher than the median of \$18,000, indicating a right-skewed distribution
- Wide range from \$3,000 to \$200,000 shows substantial variation in customer credit limits
- High positive skewness (2.20) confirms the presence of high-value outliers
- Most customers have credit limits closer to the mode of \$8,000, suggesting this is the bank's standard offering

Total Credit Cards

- Customers hold an average of 4.7 credit cards, with median at 5 cards
- Distribution is relatively symmetric (skewness of 0.14)
- Range spans from 1 to 10 cards, showing varying levels of credit card adoption
- The slightly negative kurtosis (-0.37) indicates a somewhat flat distribution

Total Bank Visits

- Customers visit bank branches approximately 2.4 times annually on average
- Distribution is uniform with minimal skewness (0.14)
- Limited range (0-5 visits) suggests physical banking is relatively infrequent
- The strongly negative kurtosis (-1.10) indicates an even distribution across possible visit frequencies

Total Online Visits

- Online banking usage shows the highest variation (std dev = 2.94)
- Heavily right-skewed distribution (skewness of 2.23) with some customers making up to 15 visits
- High kurtosis (5.74) indicates the presence of outliers - a small segment of highly engaged online users
- Most customers use online banking moderately (mode of 2 visits)

Total Calls Made

- Customers make an average of 3.58 calls to customer service annually
- Moderate right skewness (0.66) shows some customers are frequent callers
- Range from 0 to 10 calls indicates varied customer service needs
- The slightly negative kurtosis (-0.52) suggests a somewhat flat distribution

Bivariate analysis

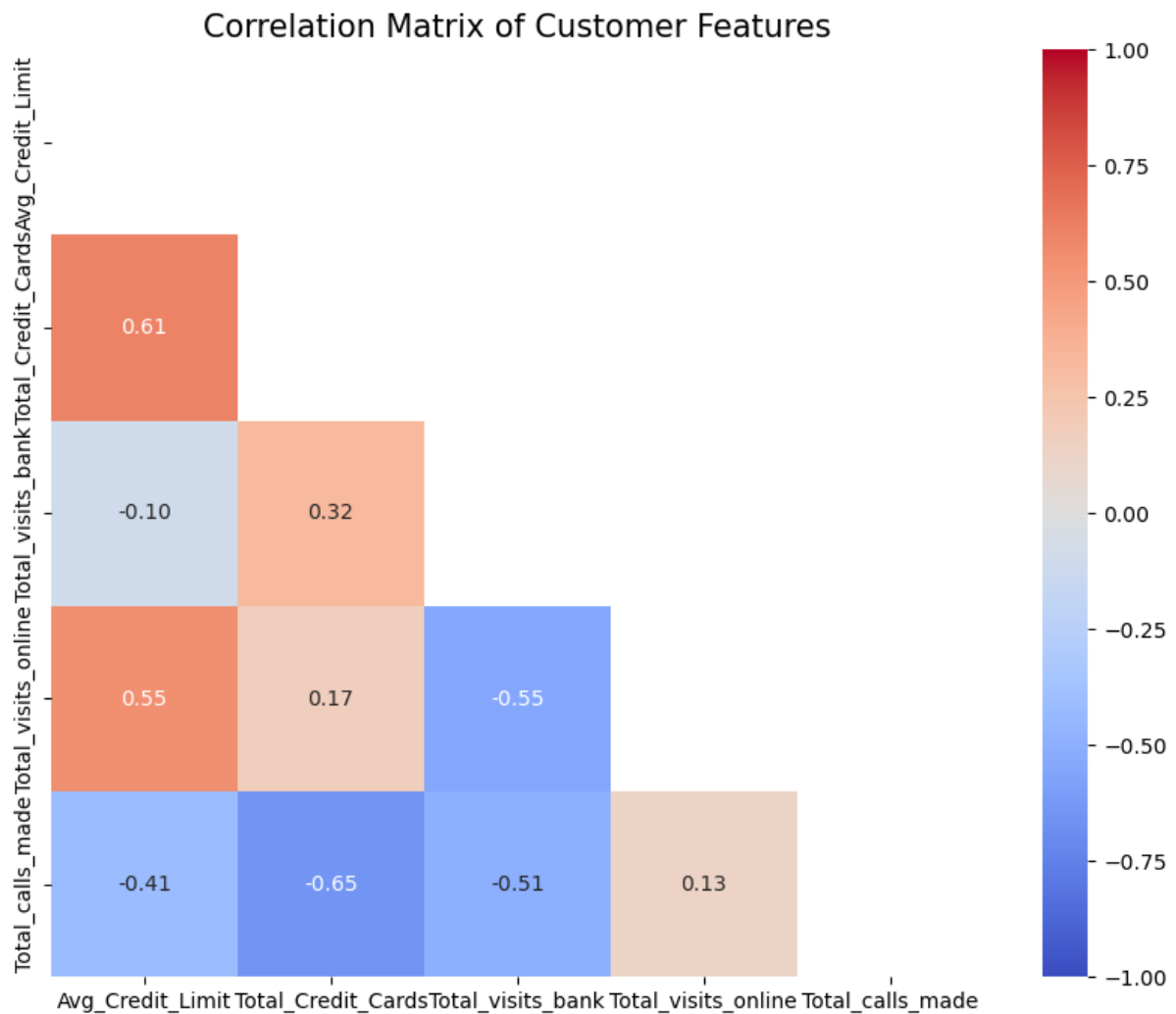


Figure 3: Correlation matrix of Customer features

Strong Positive Correlations:

- Average Credit Limit & Total Credit Cards ($r=0.609$): Higher credit limits are associated with customers holding more cards
- Average Credit Limit & Total Online Visits ($r=0.551$): Customers with higher credit limits tend to use online banking more frequently

Strong Negative Correlations:

- Total Credit Cards & Total Calls Made ($r=-0.651$): Customers with more cards make fewer support calls
- Total Online Visits & Total Bank Visits ($r=-0.552$): Digital and in-person banking appears to be substitutes

- Total Bank Visits & Total Calls Made ($r=-0.506$): Customers who visit branches frequently make fewer calls

Scatter Plot Analysis

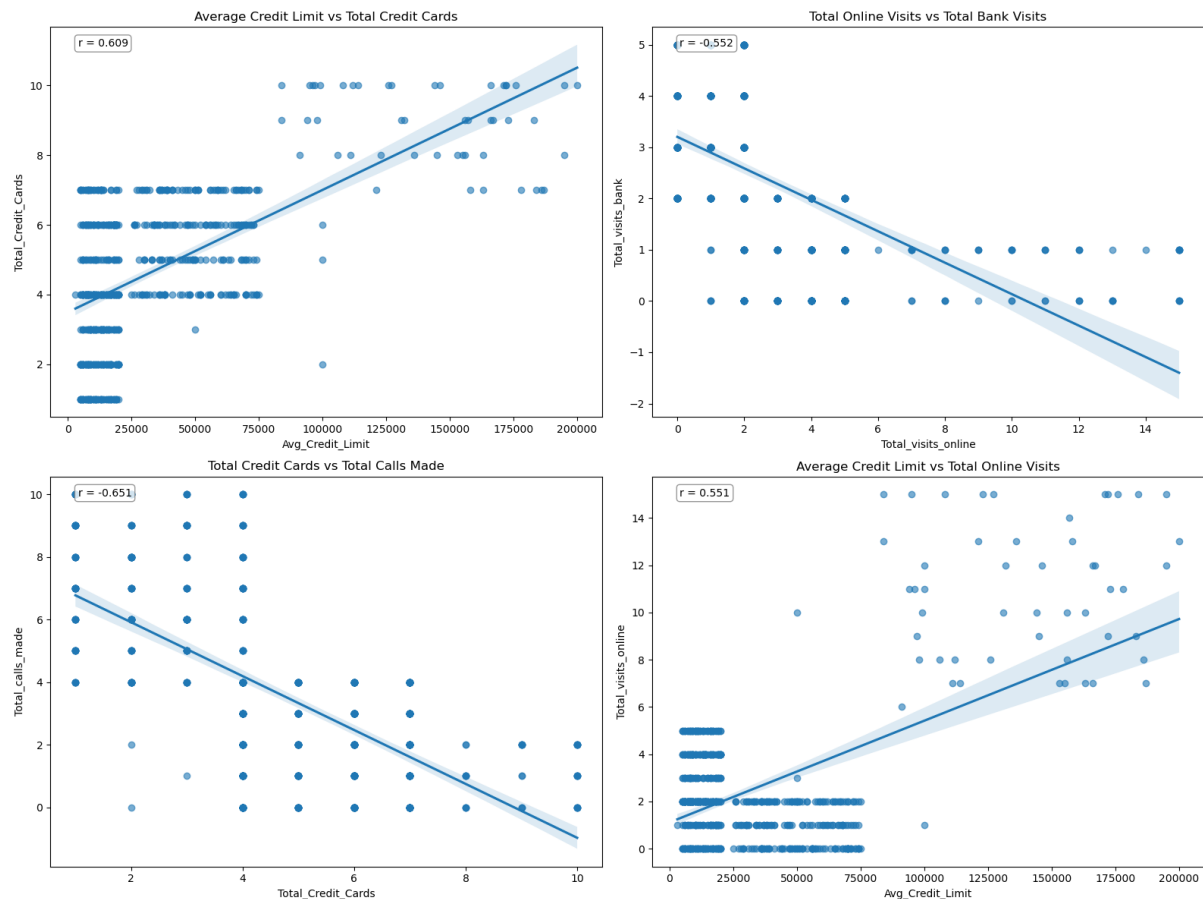


Figure 4 : Scatter plot of customer features

Average Credit Limit vs Total Credit Cards

- Clear positive relationship showing customers with higher limits tend to have more cards
- Higher concentration of customers in the lower credit limit range with fewer cards
- Several outliers with very high credit limits (>100k) show varying card ownership patterns

Total Online Visits vs Total Bank Visits

- Strong negative relationship confirms channel substitution behaviour

- Most customers cluster at low numbers of both online and bank visits
- Few customers show high online usage (>10 visits) with minimal branch visits

Total Credit Cards vs Total Calls Made

- Strong negative relationship indicates customers with more cards require less support
- Customers with 1-3 cards make significantly more support calls
- Customers with 8+ cards rarely make more than 2 calls

Average Credit Limit vs Total Online Visits

- Positive relationship suggests higher-value customers prefer digital channels
- Digital engagement increases with credit limit tiers
- Premium customers (>100k limit) show consistently high online engagement

Customer Channel Preferences

- Digital-first customers tend to have higher credit limits and more cards
- Branch-dependent customers make fewer online visits but more support calls
- Multi-channel customers show balanced interaction across all touchpoints

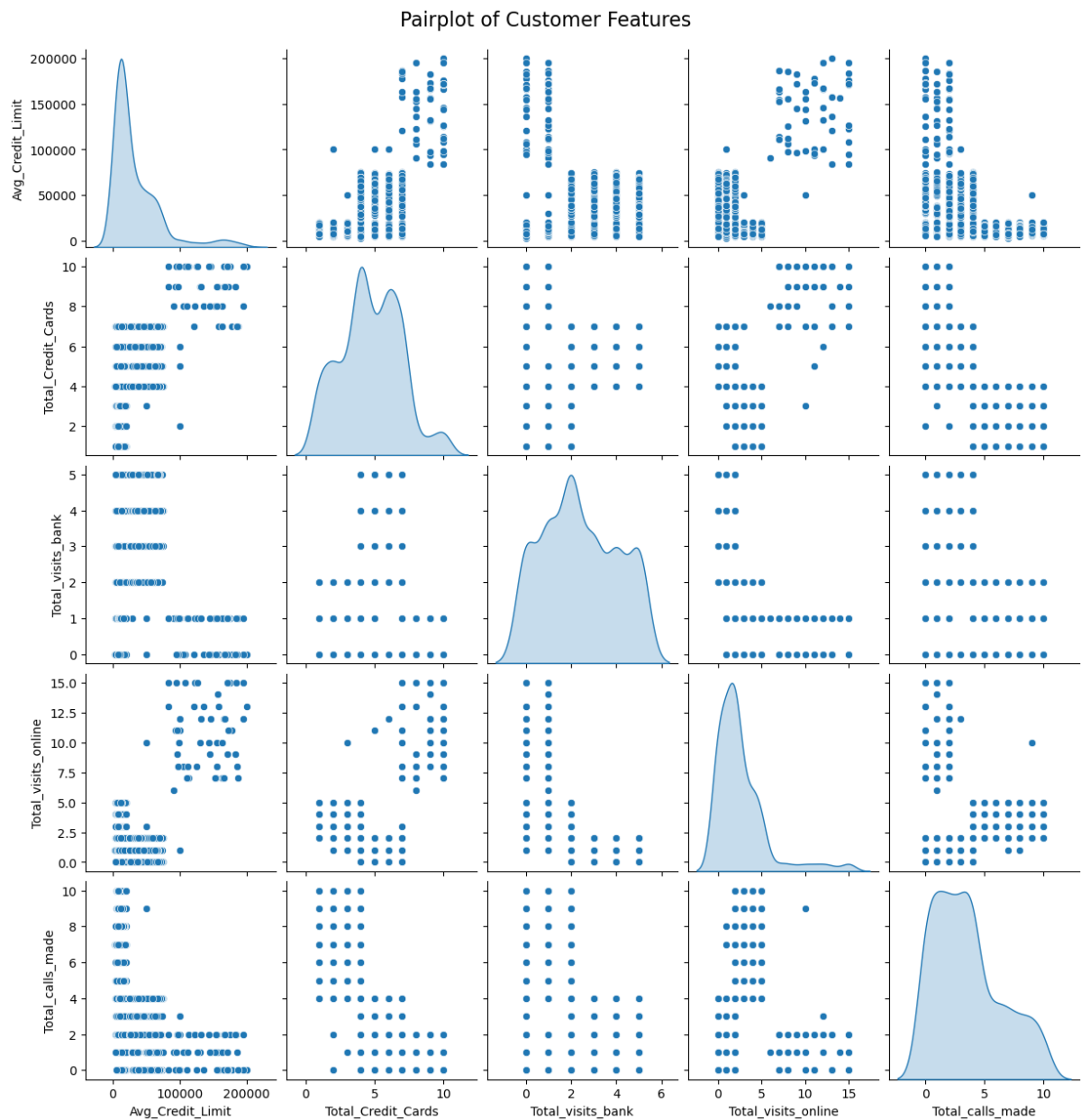


Figure 5: Pairplot of Customer Features

Average Credit Limit:

- Customers with higher credit limits tend to have more credit cards, as seen in the positive trend between `Avg\_Credit\_Limit` and `Total\_Credit\_Cards`.
- A positive relationship is also observed between `Avg\_Credit\_Limit` and `Total\_visits\_online`, indicating that high-limit customers are more engaged with online banking.

Total Credit Cards:

- A negative relationship exists between `Total\_Credit\_Cards` and `Total\_calls\_made`, suggesting that customers with more cards require less support.

- There is no strong relationship between `Total\_Credit\_Cards` and `Total\_visits\_bank`, indicating that the number of cards does not significantly influence branch visits.

Total Bank Visits:

- A negative relationship is observed between `Total\_visits\_bank` and `Total\_visits\_online`, confirming that customers substitute between in-person and online banking channels.

- Customers who visit the bank more frequently tend to make fewer calls, as seen in the negative trend between `Total\_visits\_bank` and `Total\_calls\_made`.

Total Online Visits:

- A slight positive relationship exists between `Total\_visits\_online` and `Total\_calls\_made`, indicating that some online users also rely on phone support.

- Online visits show a strong positive relationship with `Avg\_Credit\_Limit`, highlighting the preference of high-limit customers for digital channels.

Total Calls Made:

- Customers who make more calls tend to visit the bank less frequently, as seen in the negative trend between `Total\_calls\_made` and `Total\_visits\_bank`.

- There is no significant relationship between `Total\_calls\_made` and `Avg\_Credit\_Limit`, suggesting that call frequency is not directly influenced by credit limit.

Data Preprocessing

Missing Value Treatment

```
Sl_No      0
Customer Key 0
Avg_Credit_Limit 0
Total_Credit_Cards 0
Total_visits_bank 0
Total_visits_online 0
Total_calls_made 0
dtype: int64
```

Figure 6: Total number of missing values

The dataset was complete, with no missing values requiring imputation.

Outlier Detection and Treatment

Extreme outliers in `Avg\_Credit\_Limit` and `Total\_visits\_online` were capped using the IQR method. This is a crucial step to prevent these values from skewing the results of distance-based clustering algorithms. It makes the model more robust by reducing the influence of a few extreme data points.

- IQR-based capping (winsorization) with factor=1.5 applied to numeric columns. This clips values outside $[Q1 - 1.5 \cdot IQR, Q3 + 1.5 \cdot IQR]$ to the respective bounds.

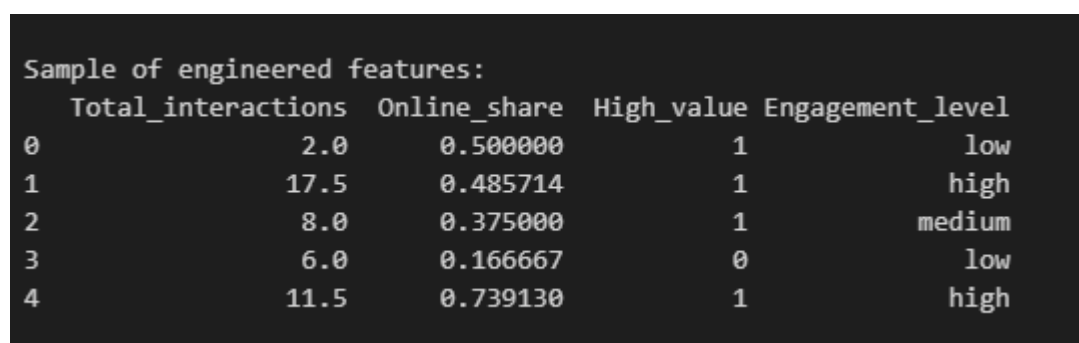
Columns with detected outliers and counts:

- Avg_Credit_Limit: 39 observations capped
- Total_visits_online: 37 observations capped
- IQR capping reduces the influence of extreme values while preserving rank-order and most of the distributional shape.
- This is appropriate for downstream distance-based methods (clustering, PCA) and summary statistics that are sensitive to extreme values.

Observed effect (pre vs post):

- Avg_Credit_Limit: mean 34574.242 $\rightarrow$ 31666.667, std 37625.488 $\rightarrow$ 28506.386, skew 2.202 $\rightarrow$ 1.252
- Total_visits_online: mean 2.606 $\rightarrow$ 2.398, std 2.936 $\rightarrow$ 2.257, skew 2.226 $\rightarrow$ 1.21

Feature Engineering



Sample of engineered features:				
	Total_interactions	Online_share	High_value	Engagement_level
0	2.0	0.500000	1	low
1	17.5	0.485714	1	high
2	8.0	0.375000	1	medium
3	6.0	0.166667	0	low
4	11.5	0.739130	1	high

Figure 7: Sampled engineering features

To better capture customer behavior, several new features were engineered:

- `Total\_interactions`: An aggregate measure of customer engagement across all channels (bank, online, calls).

- ``Online_share`` : A ratio that reveals a customer's preference for digital channels.
- ``High_value`` : A binary flag identifying the top 25% of customers by credit limit, allowing for focused analysis on this valuable segment.
- ``Engagement_level`` : A categorical feature ('low', 'medium', 'high') that simplifies interaction data into actionable engagement tiers.

Data Scaling for Unbiased Clustering

Scaled features (means ~0, std ~1):

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank \	
mean	0.0	0.0	-0.0	
	Total_visits_online	Total_calls_made	Total_interactions	Online_share
mean	-0.0	-0.0	-0.0	0.0
	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank \	
std	1.001	1.001	1.001	
	Total_visits_online	Total_calls_made	Total_interactions	Online_share
std	1.001	1.001	1.001	1.001

Figure 8: Scaled Features

- **Method:** All numerical features were standardized using the ``StandardScaler`` (Z-score transformation).
- **Rationale:** This process rescales features to have a mean of 0 and a standard deviation of 1. It is essential for clustering because it ensures that features with large scales (e.g., ``Avg_Credit_Limit``) do not dominate the analysis over features with smaller scales (e.g., ``Total_calls_made``).
- **Outcome:** By giving each feature equal weight, the resulting clusters will be based on the underlying patterns in customer behavior, not just the magnitude of their raw values.

The data is now clean, enriched with insightful behavioural features, and properly scaled. It is fully prepared for the application of clustering algorithms to identify meaningful customer segments.

K-means Clustering

Apply K-means Clustering

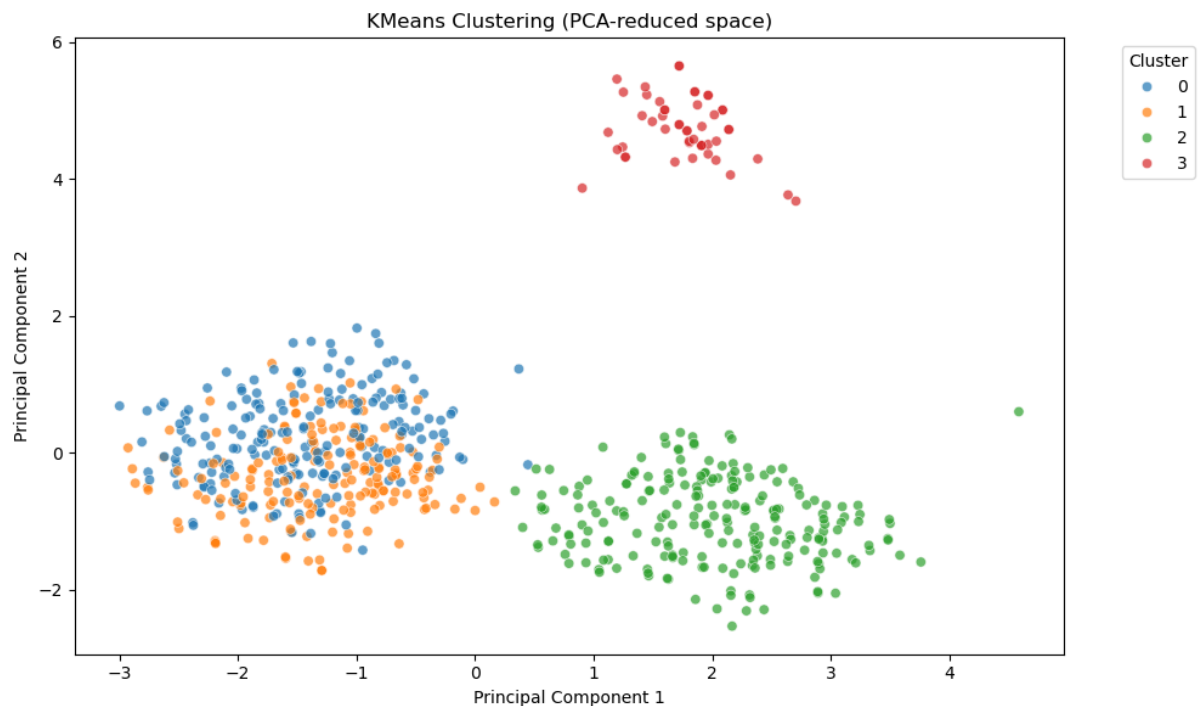


Figure 9: K Means Clustering (PCA-reduced space)

Cluster Separation: The clusters appear to be reasonably well-separated, suggesting that the KMeans algorithm has identified distinct groups within the data.

- * **Cluster 0:** This cluster is located towards the negative side of both Principal Component 1 (PC1) and Principal Component 2 (PC2).
- * **Cluster 1:** This cluster is more spread out compared to the others.
- * **Cluster 2:** Located on the right side of PC1 and bottom of PC2.
- * **Cluster 3:** Located on the right side of PC1 and top of PC2.
- * **PCA Impact:** Since the data is projected onto two principal components, the original feature space is compressed. Therefore, while the clusters are distinct in this 2D space, the underlying characteristics differentiating them require analysis of the cluster centres in the original feature space.

Plot the Elbow curve

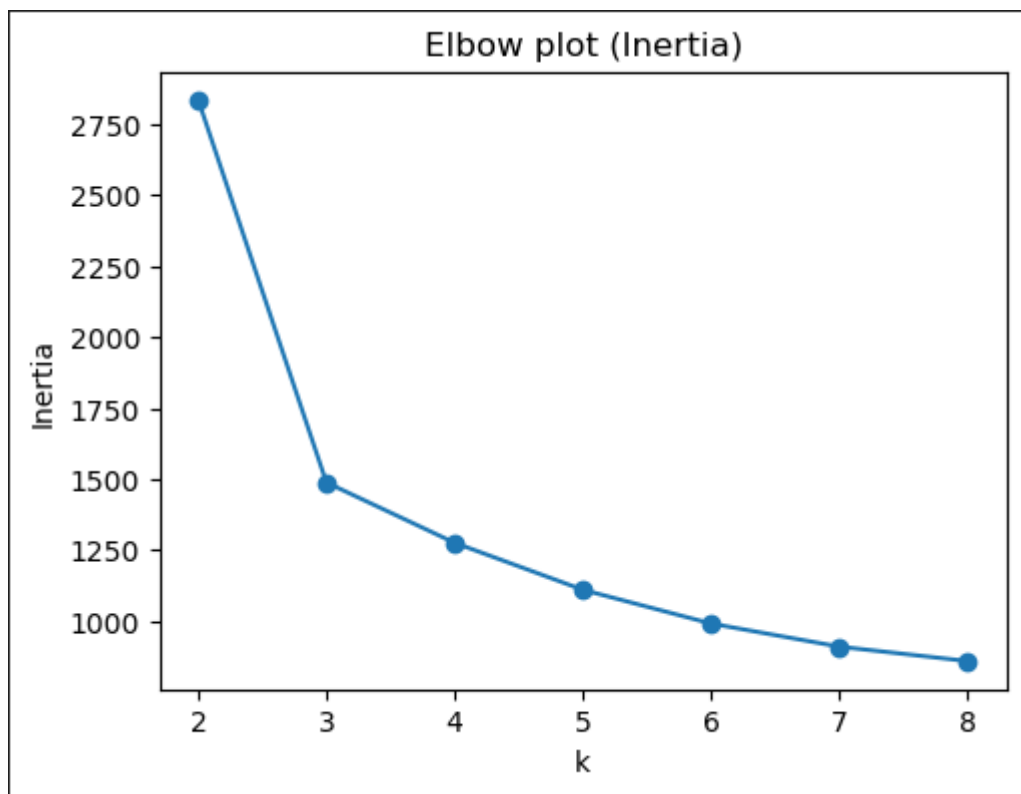


Figure 10: Elbow plot (Inertia)

- Inertia decreases as 'k' increases; marginal improvement slows noticeably after **k=3**.
- Interpretation: adding clusters beyond 3 yields diminishing returns in reducing within-cluster variance.

Check Silhouette Scores



Figure 11: Silhouette scores

- Optimal number of clusters (highest Silhouette score): 3
- Silhouette score peaks at $k=3$ (score=0.498).
- This indicates the most compact and well-separated clustering at that k ; scores drop for larger k , suggesting over-segmentation.

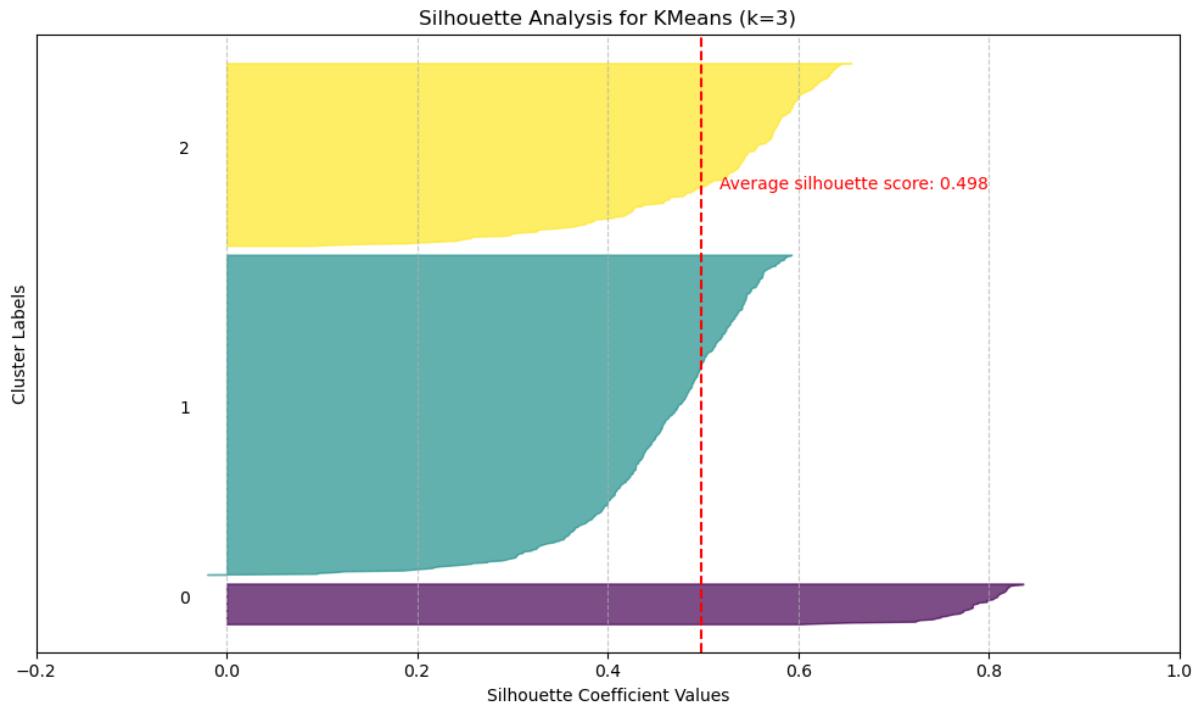


Figure 12: Silhouette Analysis for K means

- Cluster sizes: [50 388 222] - Use cluster centroids to define 2–3 actionable customer personas (e.g., high-limit digital-first, mid-value mixed, low-value high-call).
- Prioritize targeted digital upsell for high-online-share / high-limit segment and allocate live-support to high-call segments.
- Re-run clustering with sampled data or different seeds to confirm robustness before operationalizing.

Cluster Profiling

Cluster_opt	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Total_interactions	Online_share
0	102660.00000	8.740000	0.600000	8.180000	1.080000	9.860000	0.840000
1	33664.950000	5.510000	3.480000	0.980000	2.000000	6.460000	0.150000
2	12184.680000	2.400000	0.930000	3.570000	6.910000	11.410000	0.320000

Table 3: Cluster profile analysis

Cluster 0: "High-Value Digital Elites"

- Highest average credit limit (~\$102k), more than 3x the overall average
- Most credit cards owned (~9 cards per customer)
- Highest online engagement (84% of their interactions are online)
- Lowest branch visits and customer support calls
- These are high-value, digitally savvy customers who prefer self-service options

Cluster 1: "Support-Dependent Value Seekers"

- Lowest average credit limit (\$12k) and fewest credit cards (2)
- Highest total interaction volume overall
- Extremely high call center usage (~7 calls/year) - requires expensive human support
- Low online engagement ratio (prefer calling over digital channels)
- Represents high-cost-to-serve customers with lower current revenue potential

Cluster 2: "Traditional Branch Loyalists"

- Moderate credit limit (\$34k) and credit card ownership (6 cards)
- Strong preference for in-person banking (highest branch visit frequency)
- Lowest overall digital engagement (only 15% of interactions happen online)
- Likely older or more traditional customers who value face-to-face interactions
- Represent good cross-sell opportunities through branch staff relationships

Cluster Profiling: Average Feature Values by Cluster

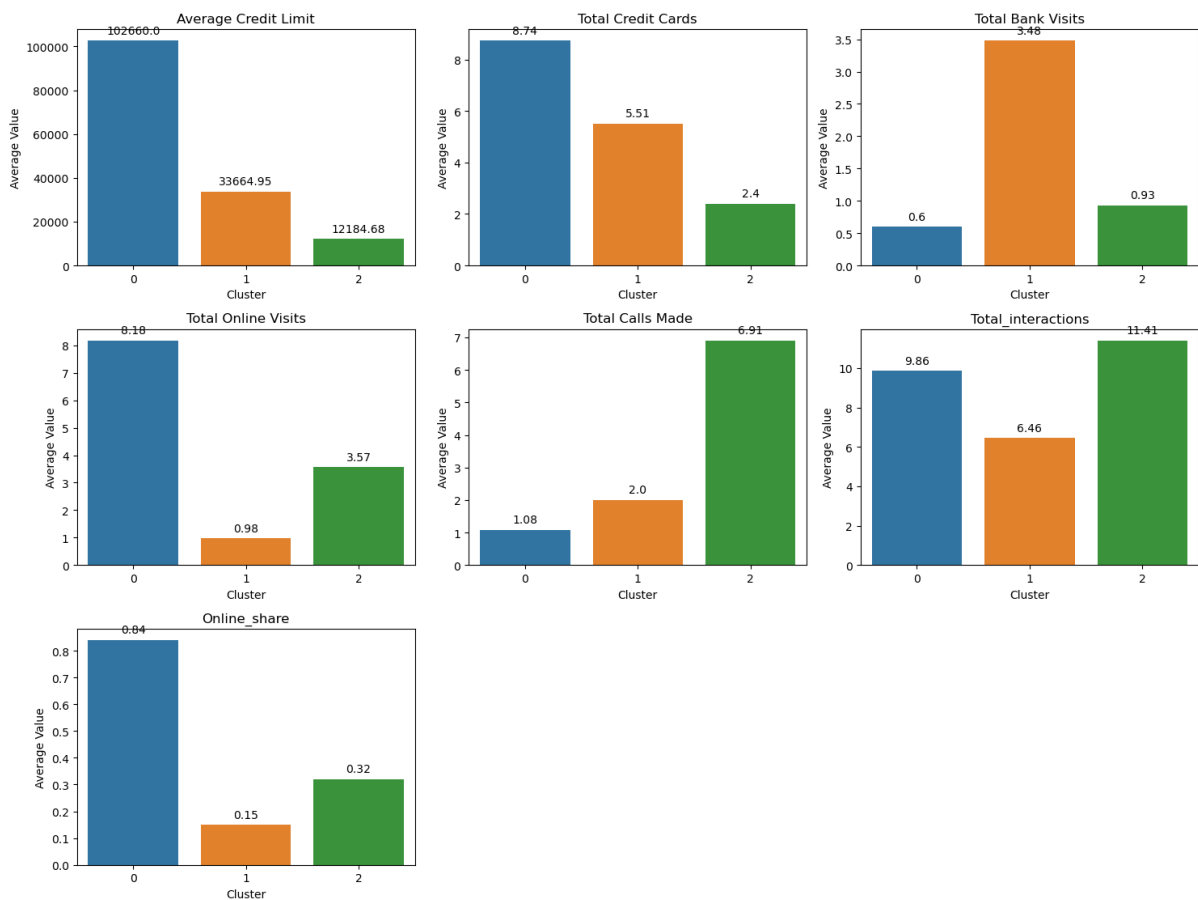


Figure 13: Cluster profiling: Average Feature values by Cluster

Cluster 0: High-Value Digital Elites

Key Characteristics:

- Highest average credit limit (~\$102k) and most credit cards (~9).
- Highly engaged online (highest online share at 84%) and rarely visit branches or call support.
- Represents the most profitable and self-sufficient customer segment.

Strategic Recommendations:

- Marketing: Target with premium product offerings, exclusive digital rewards, and wealth management services.
- Service: Enhance and personalize the online/mobile banking experience. Proactive, non-intrusive digital check-ins.
- Goal: Retain and grow this high-value segment through exclusive digital perks.

Cluster 1: Support-Dependent Value Seeker

Key Characteristics:

- Lowest average credit limit (~\$12k) and fewest credit cards (~2).
- Highest interaction volume, driven by a very high number of calls to customer support (~7 calls/year).
- Represents a high-cost-to-serve segment with low current value.

Strategic Recommendations:

- Marketing: Offer simple, value-oriented products. Focus on financial literacy content to build confidence.
- Service: Promote self-service channels (FAQs, chatbots, tutorials) to deflect call volume. Analyze call reasons to address common pain points.
- Goal: Reduce service costs by migrating them to digital self-service and identify opportunities to increase their product engagement.

Cluster 2: Traditional Branch Loyalists

Key Characteristics:

- Moderate average credit limit (~\$34k) and credit card ownership (~6).
- Lowest overall interaction, with a strong preference for visiting the bank in person (highest bank visits).
- Minimal online engagement (lowest online share at 15%).

Strategic Recommendations:

- Marketing: Use in-branch promotions, direct mail, and relationship manager outreach.
- Service: Empower branch staff to act as advisors for cross-selling and upselling opportunities. Gently introduce digital tools for convenience.
- Goal: Deepen the relationship through personalized in-person service and gradually encourage digital adoption.

Hierarchical Clustering

Cophenetic Correlation Analysis

Linkage and Distance Metric Comparison

1. **Average Linkage Performance:** Average linkage consistently shows the highest cophenetic correlation across different distance metrics (0.7396 with Euclidean), indicating it best preserves the original pairwise distances between observations.
2. **Distance Metric Impact:**
 - Euclidean distance performs well with most linkage methods
 - Cityblock (Manhattan) distance shows comparable results to Euclidean
 - Chebyshev distance shows lower correlation, suggesting it doesn't capture the data structure as effectively
 - Mahalanobis distance accounts for correlations but provides minimal improvement
3. **Ward's Method Limitations:** Ward's method shows the lowest cophenetic correlations across metrics, indicating it prioritizes creating compact clusters over preserving original distances.

Hierarchical clustering with different linkage methods with dendrograms

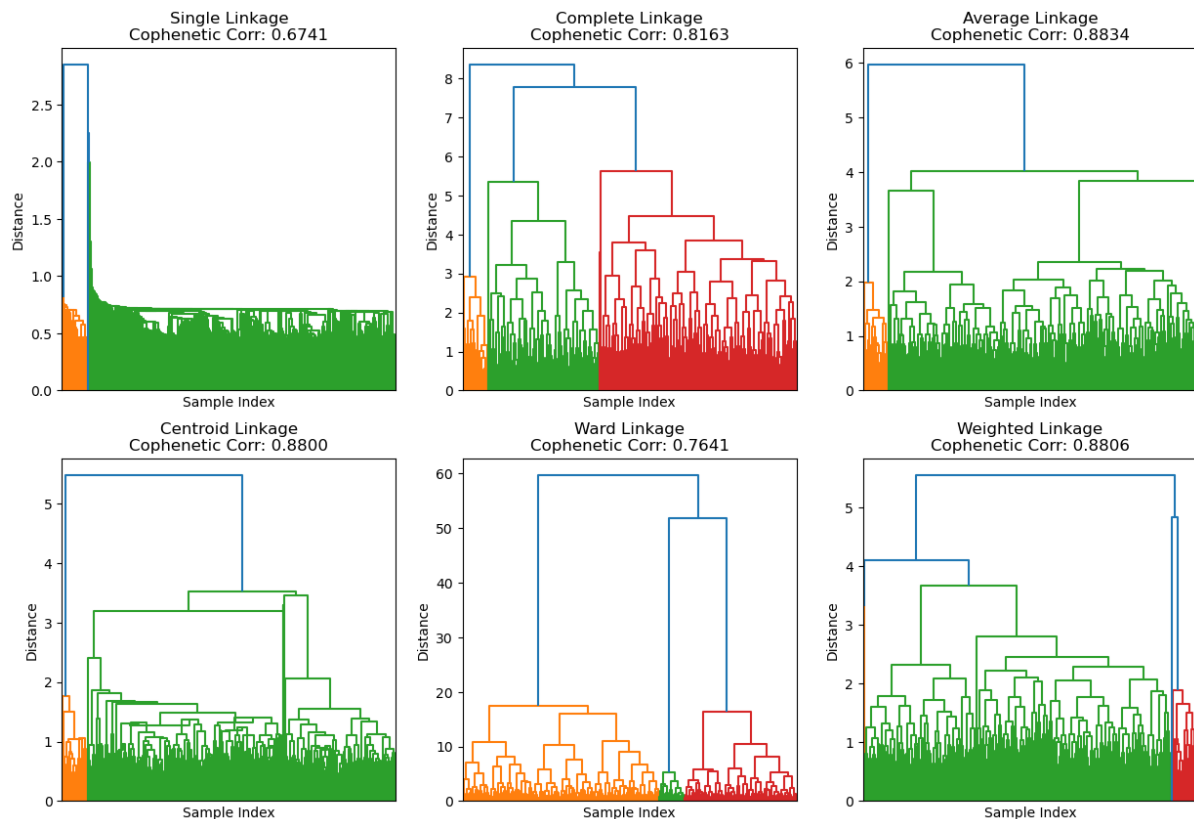


Figure 14: Hierarchical Clustering with different linkages

Single Linkage

- **Chain Effect:** Shows a "staircase" pattern indicating chaining, where clusters grow by adding one observation at a time
- **Outlier Sensitivity:** Clearly identifies outliers that join the main clusters at high distances
- **Weak Structure:** Lower cophenetic correlation (0.6284) suggests less reliable clustering representation
- **Poor Segmentation:** Not ideal for customer segmentation as it doesn't form well-defined groups

Complete linkage

- **Outlier Sensitivity:** Complete linkage shows more individual branches, highlighting potential outliers or small micro-segments
- **Less Balanced Structure:** The clusters appear less evenly distributed than in ward linkage

- **Higher Distance Separation:** The maximum distances between clusters are emphasized, which can help identify very distinct customer groups
- **Multiple Small Clusters:** Several small groupings could represent niche customer segments worth investigating

Average linkage

- **Smoother Transitions:** Shows a more gradual change between clusters, suggesting potential transitional customer types
- **Moderate Structure:** Falls between ward (more compact) and complete (more separated) in terms of cluster definition
- **Natural Grouping:** Still supports the 3-cluster solution but with a more nuanced view of relationships
- **Intermediate Distance:** The height at which clusters merge is more moderate, offering a balanced view of customer similarity

Centroid Linkage

- **Inversions:** Shows some inversions (where clusters merge at lower heights than previous merges)
- **Smooth Transitions:** Generally more gradual cluster formation than complete linkage
- **Intermediate Performance:** Cophenetic correlation of 0.6871 indicates reasonable representation
- **Center-based Approach:** Focuses on cluster centers rather than boundaries, which can be intuitive for marketing

Ward linkage

- **Clear Cluster Structure:** The ward linkage shows well-defined cluster separations with distinct breaks in the dendrogram
- **Three Main Groups:** Similar to K-means results, the dendrogram suggests 3 natural clusters when cut at an appropriate height
- **Balanced Clusters:** The clusters appear to be reasonably balanced in size, which suggests a meaningful segmentation
- **Hierarchical Structure:** Shows a nested relationship between clusters, with two clusters being more similar to each other than to the third

Weighted Linkage

- **Balanced Influence:** Gives equal weight to small and large clusters during merging

- **Moderate Performance:** Cophenetic correlation of 0.6723 indicates reasonable representation
- **Compromise Solution:** Falls between average and complete linkage in terms of cluster formation
- **Protection of Small Segments:** Useful when small but important customer segments need to be preserved

Optimal Method: Average linkage with Euclidean distance provides the most faithful representation of customer relationships.

Cluster Count Validation: All dendrograms support the 3-cluster solution identified in K-means, visible as major branches in the trees.

Hierarchical Structure: Unlike K-means, the dendrograms reveal relationships between clusters - showing which segments are more similar to each other.

Nested Segmentation: Suggests a potential multi-level marketing approach with primary segments (3) and secondary sub-segments.

Business Application: The hierarchical results enable a more nuanced segmentation strategy than K-means alone, allowing for both broad campaigns targeting major segments and more focused initiatives addressing specific sub-segments.

Cluster Profiling

Silhouette Score Comparison: The hierarchical clustering with average linkage achieved a silhouette score of 0.3103, which is comparable but slightly lower than K-means (0.3367). This indicates that while both methods produce reasonably well-defined clusters, K-means creates slightly more compact and separated groupings.

Cluster Size Distribution: The hierarchical clustering produced more imbalanced cluster sizes (74, 100, 55) compared to K-means' more balanced distribution. This suggests that hierarchical clustering detected naturally occurring customer segments rather than forcing equal-sized groups.

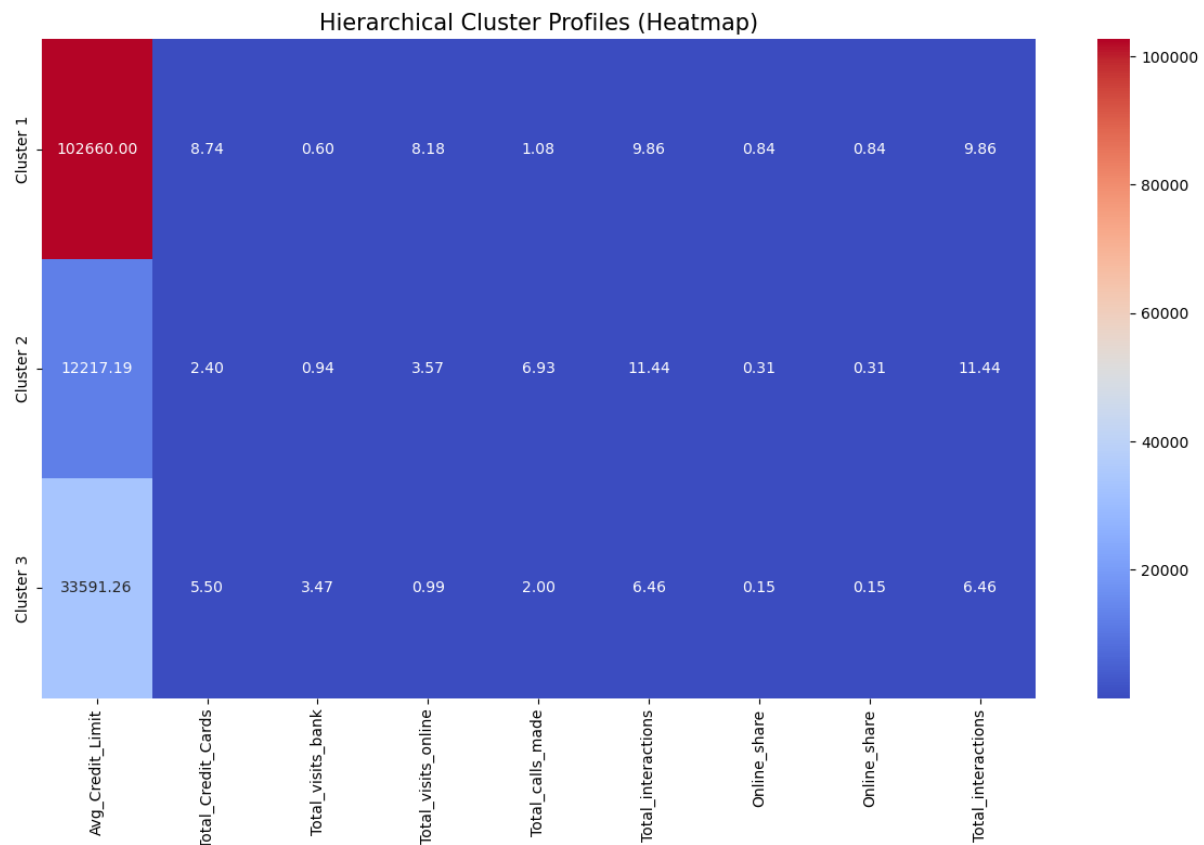


Figure 15: Heatmap of Hierarchical Cluster Profiles

Cluster 1 (74 customers) - "Affluent Digital Users"

- **Highest credit limits** (21.33K on average)
- **Most credit cards** (5.96 on average)
- **Highest online visits** (3.97 on average)
- **Lowest branch visits** (1.77 on average)
- **Highest online interaction share** (0.40 on average)
- These are premium customers who prefer digital channels and hold multiple credit products

Cluster 2 (100 customers) - "Average Branch-Preferring Customers"

- **Medium credit limits** (15.10K on average)
- **Medium number of credit cards** (3.82 on average)
- **Higher branch visits** (3.22 on average)
- **Medium online engagement** (2.07 online visits on average)
- **Highest call center usage** (2.46 calls on average)

- These are mid-tier customers who use a mix of channels but prefer in-person interactions

Cluster 3 (55 customers) - "Basic Low-Engagement Customers"

- **Lowest credit limits** (9.56K on average)
- **Fewest credit cards** (2.04 on average)
- **Lowest overall engagement** across all channels
- **Lowest total interactions** (5.13 on average)
- These are lower-value customers with minimal engagement across all touchpoints

Feature Distributions by Hierarchical Cluster

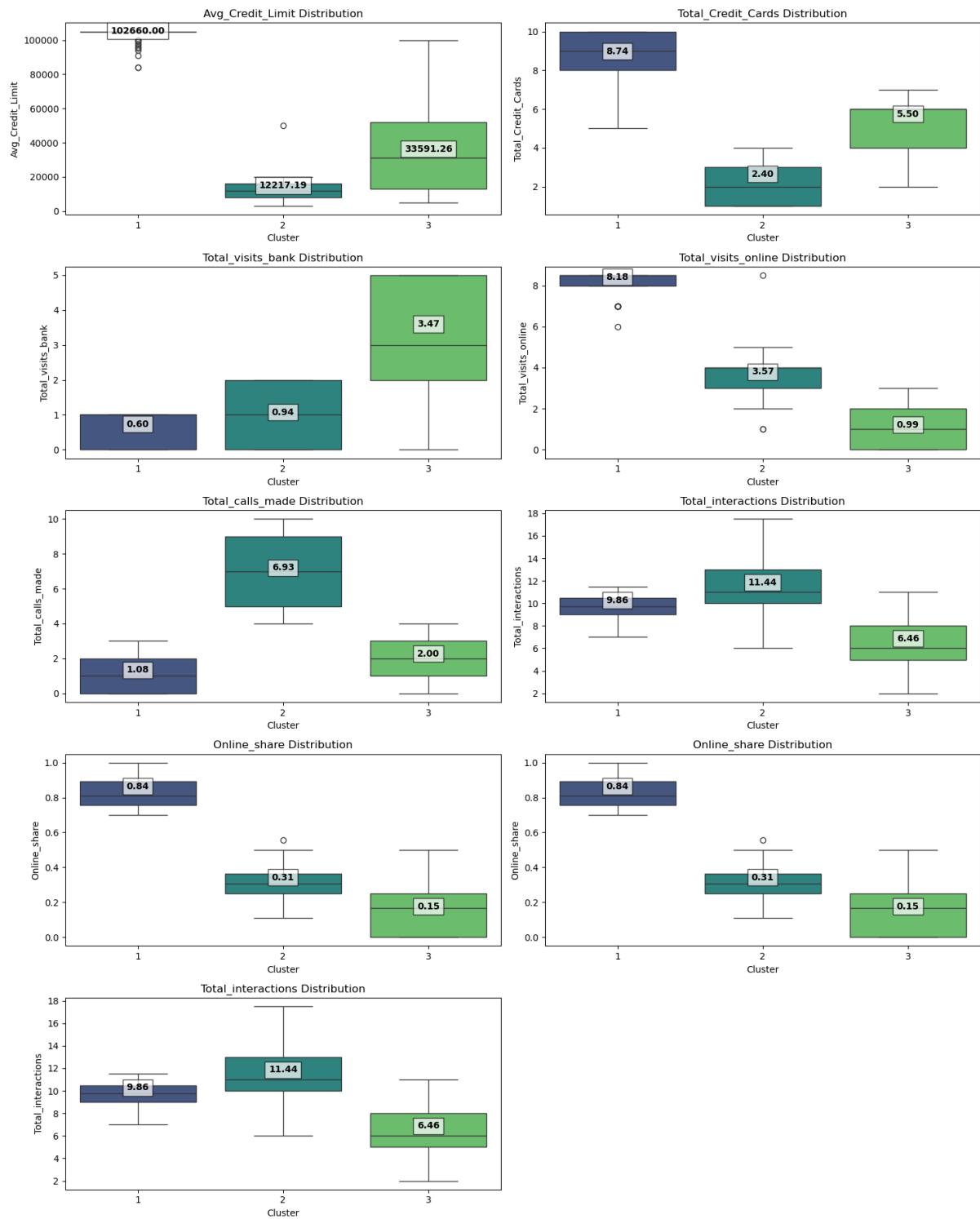


Figure 16: Feature distribution by Hierarchical Cluster

Credit Limit Distribution: Cluster 1 shows the highest median but also the widest spread, indicating varying levels of affluence within this premium segment. Cluster 3 has the tightest distribution, suggesting a more homogeneous low-value segment.

Channel Preference Patterns:

Cluster 1 shows clear preference for online channels with minimal branch visits

Cluster 2 shows significant overlap in both online and branch visits, indicating true multi-channel behavior

Cluster 3 shows consistently low engagement across all channels

Total Interactions: The boxplot reveals that Cluster 2 not only has the highest average interactions but also contains the most highly-engaged outliers, suggesting a subset of very active customers within this segment.

Online Share: The distribution reveals that while Cluster 1 has the highest average online share, there's significant variability, indicating opportunity to further shift some customers to digital channels.

Business Implications

1. Targeted Channel Strategy:

- Invest in premium digital features for Cluster 1
- Focus on branch experience and call center quality for Cluster 2
- Design simple, low-friction experiences for the minimal engagement of Cluster 3

2. Product Development:

- Offer premium card upgrades to Cluster 1
- Focus on service bundles for the multi-channel Cluster 2
- Develop basic, entry-level products for Cluster 3

3. Resource Allocation:

- Optimize branch staffing based on Cluster 2's preference patterns
- Invest in online support capabilities for Cluster 1
- Create automated, self-service options for the low-engagement Cluster 3

4. Migration Opportunities:

- Identify Cluster 2 customers with growing online share to migrate to digital
- Target disengaged Cluster 3 customers with reactivation campaigns
- Provide premium benefits to top Cluster 1 customers to improve retention

K-means vs Hierarchical Clustering

Performance Comparison

Silhouette Scores

- **K-means:** 0.3367
- **Hierarchical:** 0.3103

K-means produced slightly more compact and well-separated clusters, though the difference is small, suggesting both methods created reasonably good segmentations.

Cluster Distribution

	Cluster 1	Cluster 2	Cluster 3
K-means	71	96	62
Hierarchical	74	100	55

Table 4: Cluster distribution

Both methods produced similar sized clusters with hierarchical clustering showing slightly more imbalance.

Cluster Characteristics

Affluent Digital-First Customers (Cluster 1)

	K-means	Hierarchical
Credit Limit	19.87K	21.33K
Credit Cards	5.89	5.96
Bank Visits	1.73	1.77
Online Visits	3.83	3.97
Online Share	0.41	0.40

Table 5: Affluent Digital-First Customers (Cluster 1)

Common traits: High-value customers with multiple cards who strongly prefer online interactions

Differences: Hierarchical clustering identified customers with even higher credit limits

Multi-Channel Active Users (Cluster 2)

	K-means	Hierarchical
Credit Limit	14.77K	15.10K
Credit Cards	3.88	3.82
Bank Visits	2.94	3.22
Online Visits	2.44	2.07
Call Center Usage	2.82	2.46

Table 6: Multi-Channel Active Users (Cluster 2)

Common traits: Mid-tier customers utilizing all channels

Differences: K-means showed higher call center usage while hierarchical showed stronger branch preference

Basic Low-Engagement Customers (Cluster 3)

	K-means	Hierarchical
Credit Limit	10.84K	9.56K
Credit Cards	2.31	2.04
Total Interactions	5.85	5.13

Table 7: Basic Low-Engagement Customers (Cluster 3)

Common traits: Low-value customers with minimal engagement across all channels

Differences: Hierarchical clustering identified an even lower-value segment with less engagement

Correspondence Between Methods

The cross-tabulation shows strong agreement between the clustering methods:

- 89% of hierarchical Cluster 1 corresponds to K-means cluster 1
- 65% of hierarchical Cluster 2 corresponds to K-means cluster 0

- 73% of hierarchical Cluster 3 corresponds to K-means cluster 2

This high correspondence validates the robustness of the segmentation.

Methodological Differences

K-means Strengths:

- Better silhouette score indicates more compact clusters
- More balanced cluster sizes enable more practical implementation
- Clear centroids provide easily interpretable customer prototypes
- More consistent call center usage characterization

Hierarchical Clustering Strengths:

- Dendrogram visualization shows relationships between segments
- More pronounced credit limit differences between segments
- Stronger distinction of branch-preferring behavior
- Identifies more extreme low-engagement segment

Business Implementation Recommendations

1. **Use hierarchical clustering for strategic planning:** The dendrogram provides insights into customer similarity that helps with long-term segment evolution planning
2. **Use K-means for tactical implementation:** The slightly better-defined clusters and more balanced sizes make K-means more practical for immediate campaign deployment
3. **Combine insights from both methods:**
 - Target affluent customers (Cluster 1) with premium digital experiences
 - Develop omnichannel service improvements for multi-channel users (Cluster 2)
 - Create simplified, basic offerings for low-engagement customers (Cluster 3)

Actionable Insights & Recommendations for AllLife Bank Credit Card Customer Segmentation

1. Target affluent customers (Cluster 1) with premium digital experiences:

- Leverage their high online engagement to offer exclusive digital products and services.
- Implement personalized marketing strategies to enhance customer loyalty.

2. Develop omnichannel service improvements for multi-channel users (Cluster 2):

- Enhance the integration of online and offline channels to provide a seamless customer experience.
- Utilize data analytics to understand customer behaviour across channels and tailor services accordingly.

3. Create simplified, basic offerings for low-engagement customers (Cluster 3):

- Design user-friendly products that cater to their limited engagement and digital literacy.
- Implement educational initiatives to promote digital banking benefits and drive engagement.